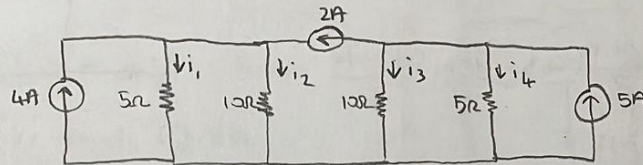


Question 10



Node 10

$$i_1 = 4A$$

Node 20

$$\text{KCL } \Rightarrow i_1 = i_2 + 2$$

$$i_2 = 2A$$

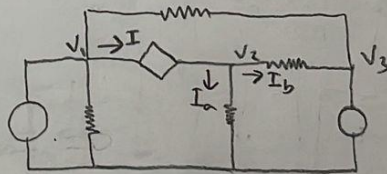
Node 30

$$\text{KCL } \Rightarrow i_2 = i_3 = 2A$$

Node 40

$$i_3 = i_4 - 5 \quad i_4 = 7A$$

Question 20



$$\text{for } V_1 \rightarrow -2 + V_1 + \frac{V_1 - V_3}{2} + I = 0$$

$$\text{for } V_2 \rightarrow V_2 = V_0$$

$$\text{for } I \rightarrow I_a = \frac{V_2}{4}, I_b = \frac{V_2 - V_3}{8}$$

$$\text{for } V_3 \rightarrow V_3 = 13V$$

$$\text{for } V_1 - V_2 \rightarrow V_1 - V_2 = 2V_0 \\ V_1 = 3V_2$$

$$(1) -2 + V_1 + \frac{V_1 - V_3}{2} + \frac{3V_2 - V_3}{8} = 0$$

$$(2) V_1 = 3V_2$$

$$-16 + 12V_1 - 5V_3 + 3V_2 = 0$$

$$13V_1 = 5V_3 + 16$$

$$V_1 = \frac{81}{13}$$

$$V_2 = \frac{81}{39}$$

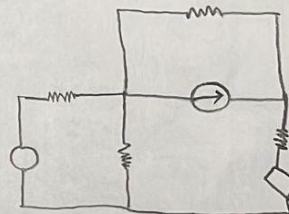
$$V_3 = 13$$

Question 3

$$\text{for } V_1 \rightarrow \frac{V_1 - 12}{2} + 3 + \frac{V_1}{4} + \frac{V_1 - V_2}{8} = 0$$

$$V_0 = V_1 - 12$$

$$\text{for } V_2 \rightarrow \frac{V_2 - V_1}{8} - 3 + \frac{V_2 + 5V_0}{1} = 0$$



$$\textcircled{1} \quad 7V_1 - V_2 = +24$$

$$\textcircled{2} \quad 9V_2 - V_1 + 40V_0 = 24$$

$$9V_2 - 39V_1 - 12 \cdot 40 = 24$$

$$7V_1 - V_2 = +24$$

$$9V_2 + 39V_1 = 12(42)$$

$$102V_1 = 12(60)$$

$$V_1 = \frac{120}{17} = 7,06 \text{ V}$$

$$V_2 = 25,42$$

Question 4

$$3i_1 - 2i_2 = -20$$

$$-2i_1 + 4i_2 - 2i_3 = 30$$

$$-2i_2 + 3i_3 = 10$$

$$\begin{bmatrix} 3 & -2 & 0 \\ -2 & 4 & -2 \\ 0 & -2 & 6 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} -20 \\ 30 \\ 10 \end{bmatrix}$$

find D

$$\begin{bmatrix} 3 & -2 & 0 & 3 & -2 \\ -2 & 4 & -2 & -2 & 4 \\ 0 & -2 & 6 & 0 & -2 \end{bmatrix}$$

$$-(0 + 12 + 24) + (72 + 0 + 0)$$

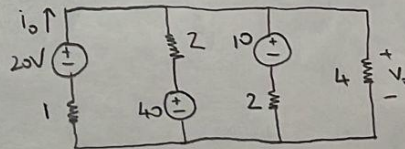
$$D = 36$$

$$i_2 = 10 \text{ A}$$

$$i_0 = 0 \text{ A}$$

$$V_0 = 20 \text{ V}$$

$$i_3 = 5 \text{ A}$$



find D1

$$\begin{bmatrix} -20 & -2 & 0 & -20 & -2 \\ 30 & 4 & -2 & 30 & 4 \\ 10 & -2 & 6 & 10 & -2 \end{bmatrix}$$

$$D_1 = 0$$

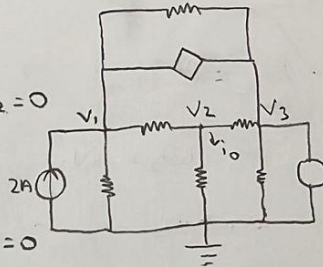
Question 5

$$i_o = \frac{V_2}{4}$$

$$\text{for } V_1 \rightarrow -2 + \frac{V_1}{2} + \frac{V_1 - V_3}{4} + 3i_o + V_1 - V_2 = 0$$

$$\text{for } V_2 \rightarrow V_2 - V_1 + V_2 - V_3 + i_o = 0$$

$$\text{for } V_3 \rightarrow -4 + \frac{V_3}{2} + V_3 - V_2 - 3i_o + \frac{V_3 - V_1}{4} = 0$$



$$\textcircled{1} \quad 7V_1 - V_2 - V_3 = 8 \quad \text{for } D_1$$

$$\textcircled{2} \quad -4V_1 + 9V_2 - 4V_3 = 0$$

$$\textcircled{3} \quad -V_1 - 7V_2 + 7V_3 = 16$$

$$\begin{bmatrix} 7 & -1 & -1 \\ -4 & 9 & -4 \\ -1 & -7 & 7 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 0 \\ 16 \end{bmatrix}$$

$$D = 176$$

for D_1

$$D_1 = 488$$

$$V_1 = \frac{488}{176} = 2.77V$$

for D_3

$$D_3 = 1240$$

$$V_3 = \frac{1240}{176} = 7.05V$$

$$\text{for } D_2$$

$$D_2 = 768$$

$$V_2 = \frac{768}{176} = 4.36V$$

Question 6

$$\text{node } \textcircled{1} \quad \frac{V_1 - V_2}{40} = \frac{100 - V_1}{10} + \frac{4V_0 - V_1}{20}$$

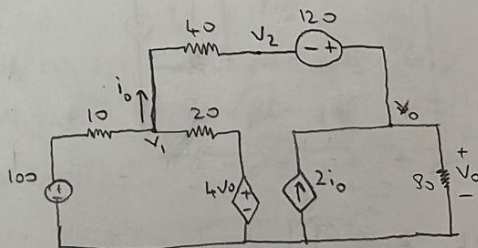
$$V_0 = 120 + V_2$$

$$\text{node } \textcircled{2} \quad 7V_1 - 9V_0 = 280$$

$$\text{node } \textcircled{2} \quad I_0 + 2I_0 = \frac{V_0}{80}$$

$$3\left(\frac{V_1 + 120 + V_2}{40}\right) = \frac{V_0}{80}$$

$$6V_1 - 7V_2 = -720$$



calculate

$$\begin{bmatrix} 7 & -9 \\ 6 & -7 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 280 \\ -720 \end{bmatrix}$$

$$\Delta = 5$$

$$V_1 = -1688$$

$$V_0 = -1344V$$

$$I_0 = -9.6A$$

Question 70

$$(1) \quad 1 + 2V_0 = \frac{V_1}{4} + \frac{V_2}{1} + \frac{V_1 - V_3}{1} \quad V_0 = V_1 - V_3$$

$$(2) \quad 4 = -3V_1 + 4V_2 + 4V_3 \quad 2V_0 + \frac{V_3}{4} = V_1 - V_3 \quad \frac{10 - V_3}{2}$$

$$(3) \quad 20 = 4V_1 - V_3$$

$$(4) \quad V_2 = V_1 + V_3$$

$$\underline{V_1 = 4,97 \text{ V}} \quad \underline{V_2 = 4,85 \text{ V}} \quad \underline{V_3 = -0,12 \text{ V}}$$

Question 80

$$\text{for } i_1 \rightarrow 12i_1 - 2i_2 = 6$$

$$\text{for } i_2 \rightarrow 7i_2 - 2i_1 - i_3 = -8$$

$$\text{for } i_3 \rightarrow 6i_3 - i_2 = 2$$

$$\begin{bmatrix} 12 & -2 & 0 \\ -2 & 7 & -1 \\ 0 & -1 & 6 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 6 \\ -8 \\ 2 \end{bmatrix}$$

for Dx

$$\begin{array}{ccccccc} 12 & -2 & 0 & 12 & -2 & & \\ -2 & 7 & -1 & -2 & 7 & & \\ 0 & -1 & 6 & 0 & -1 & & \\ 6 & -2 & 0 & 6 & -2 & & \\ -8 & 7 & -1 & -8 & 7 & & \\ 2 & -1 & 6 & 2 & -1 & & \end{array}$$

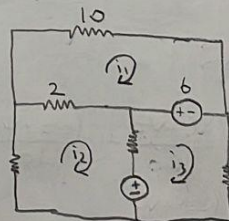
$$D = 492$$

$$36.7 + 4 - (6 + 96)$$

$$i_1 = \frac{154}{492} = 0,31 \text{ A}$$

$$i_2 = -1,12$$

$$i_3 = 0,15$$



Question 9

$$i_1 \rightarrow 40i_1 - 10i_2 = -120$$

$$i_2 \rightarrow -10i_1 + 50i_2 - 10i_3 = 0$$

$$i_3 \rightarrow -10i_2 + 40i_3 = 120$$

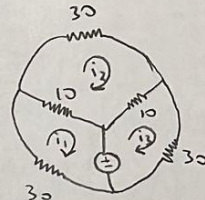
$$\begin{bmatrix} 40 & -10 & 0 \\ -10 & 50 & -10 \\ 0 & -10 & 40 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} -120 \\ 0 \\ 120 \end{bmatrix}$$

for D

$$\begin{bmatrix} 40 & -10 & 0 & 40 & -10 \\ -10 & 50 & -10 & -10 & 50 \\ 0 & -10 & 40 & 0 & -10 \end{bmatrix}$$

$$80000 - (4000 + 4000) = 72000$$

$$\boxed{i_1 = -3} \quad \boxed{i_2 = 0} \quad \boxed{i_3 = 3}$$



or D x y

$$\begin{bmatrix} -120 & -10 & 0 & -120 & -10 \\ 0 & 50 & -10 & 0 & 50 \\ 120 & -10 & 40 & 120 & -10 \end{bmatrix}$$

$$-240000 + 12000 - (-12000)$$

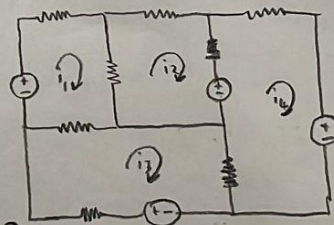
Question 10

$$i_1 \rightarrow i_1 R_1 + (i_1 - i_2) R_4 + (i_1 - i_3) R_6 - V_1 = 0$$

$$i_2 \rightarrow (i_2 - i_1) R_4 + i_2 R_2 + (i_2 - i_4) R_5 + V_2 = 0$$

$$i_3 \rightarrow i_3 R_7 + (i_3 - i_1) R_6 + (i_3 - i_4) R_8 - V_3 = 0$$

$$i_4 \rightarrow (i_4 - i_3) R_8 - V_2 + (i_4 - i_2) R_5 - i_4 R_3 + V_4 = 0$$



Question 11

$$i_1 \rightarrow -6 + 9i_1 - 3i_2 - 4i_3 = 0$$

$$i_2 \rightarrow 8i_2 - 3i_1 - 4i_3 = 0$$

$$i_3 \rightarrow 6i_3 - 4i_1 - i_4 - 2 = 0$$

$$i_4 \rightarrow 2i_4 - i_3 + 3 = 0$$

